PHISHING ANALYSIS & AWARENESS CAMPAIGN

Cybersecurity Project 02

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Project Overview:

This project examines five suspicious and phishing email samples to identify common warning signs, understand the techniques used by attackers, and learn how users can protect themselves from phishing attempts.
The analysis focuses on email sender details, authentication results, suspicious links, and social engineering techniques.

1. Introduction

Phishing is a type of cyberattack in which attackers send fake emails or messages to trick people into sharing personal information, passwords, or financial details. These emails often look like they come from trusted companies or services. Phishing emails may use fake account warnings, delivery problems, payment requests, or attractive offers to convince people to click suspicious links.
In this project, five email samples were selected and examined to understand how phishing and suspicious emails work. The samples include messages pretending to represent Microsoft, DHL, and Amazon, along with solar panel advertising and a dating-related email.
The analysis focuses on identifying warning signs in the sender's email address, email headers, website links, and message content. The purpose is to understand how these emails try to influence recipients and how similar threats can be recognized in everyday situations.

2. Project Objectives

The main objectives of this project are:

1. To examine five suspicious or phishing email samples.
2. To identify differences between the displayed sender name and the actual sender email address.
3. To check email authentication results using SPF, DKIM, and DMARC.
4. To inspect suspicious URLs safely without opening them.
5. To investigate sender domains using domain lookup tools.
6. To identify social engineering techniques such as urgency, fear, impersonation, and curiosity.
7. To compare common warning signs across the five samples.
8. To create awareness materials that help employees recognize and avoid phishing attempts.

3. Methodology and Tools Used

3.1 Collection of Email Samples

Five suspicious email samples were collected from publicly available sources, including GitHub repositories and the Phishing Pot email collection. The selected emails represented different types of suspicious messages, such as account security warnings, delivery scams, promotional messages, and dating-related emails. The original email content and headers were examined to identify possible phishing indicators.

3.2 Email Header Analysis

The email headers were inspected to identify the sender's address, sending IP address, and email authentication results. Three authentication mechanisms were examined:

- **SPF** (Sender Policy Framework): Checks whether the sending server is authorized to send emails for a domain.
- **DKIM** (DomainKeys Identified Mail): Uses a digital signature to help verify that an email was authorized by the signing domain and was not altered after signing.
- **DMARC** (Domain-based Message Authentication, Reporting and Conformance): Checks whether SPF or DKIM authentication aligns with the domain shown in the From address.

The results were recorded for each email to identify authentication failures and inconsistencies.

3.3 Domain and URL Investigation

ICANN Lookup was used to check available domain registration information, including registration dates. In cases where registration details were unavailable, this was recorded in the analysis.
Suspicious URLs were extracted from the email content and HTML source code without opening them directly.
VirusTotal was used to check whether security vendors had identified the submitted URLs as suspicious or malicious. The detection results were recorded as displayed.

3.4 Social Engineering Analysis

The content of each email was reviewed to identify techniques used to influence the recipient.
These included impersonating trusted companies, creating urgency, using fear of account suspension, and encouraging users to click links through attractive offers or personal messages.

3.5 Evidence Collection

Screenshots were saved during the investigation to document email headers, suspicious links, ICANN Lookup results, and VirusTotal scan results. The screenshots were organized by sample number and filename so they could be referenced in the final report.

4. Analysis of Phishing Email Samples

The five email samples use different tactics, including fear, urgency, impersonation, and curiosity. The following analyses highlight the warning signs found in each sample.

4.1 Sample 1 — Microsoft Account Security Alert

4.1.1 Email Overview

The first sample is a phishing email that pretends to come from Microsoft's account security team. It warns the recipient about unusual sign-in activity and asks them to review their account immediately.

Table 1: Sample_1 Email Detail

Field	Information
Display Name	Microsoft Account Team
Sender Address	noreply@microsoftonline-verify.com
Subject	[Action Required] Unusual sign-in activity on your account
Date	6 February 2026
Sender IP	178.238.225.91

4.1.2 Sender and Domain Analysis

The sender uses the name "**Microsoft Account Team**" to make the email appear official. However, the actual email address uses microsoftonline-verify.com, which is not Microsoft's official domain.
The domain includes the words "**microsoft**" and "**verify**" to make it look trustworthy. This is an example of **brand impersonation** using a lookalike domain.
The domain was also checked using ICANN Lookup, but no registration record was found. Therefore, its registration date and age could not be confirmed.

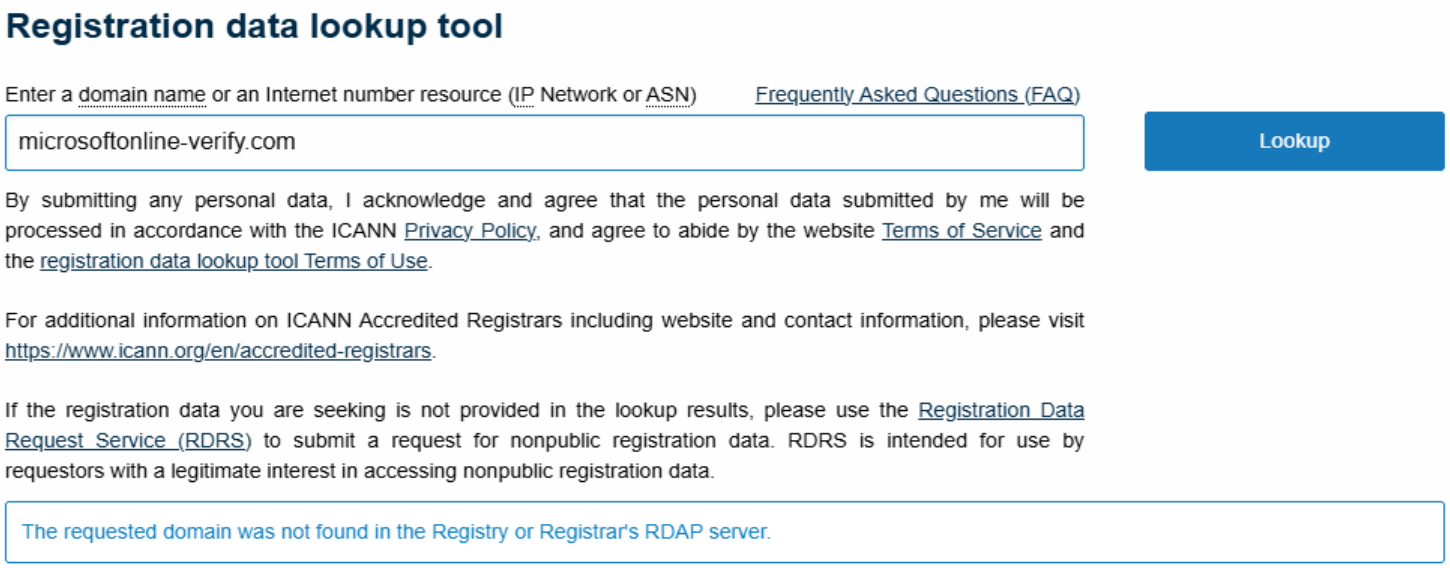


Figure 1: Microsoft Domain Check

4.1.3 Email Authentication Results

The email header showed the following results:

- **SPF: FAIL** — The sending server was not authorized according to the SPF check.
- **DKIM: FAIL** — The email did not pass the digital signature verification.
- **DMARC: FAIL** — The email failed the domain authentication policy check.

These failures increase suspicion because the email could not successfully authenticate its claimed sending domain.



Figure 2: Microsoft Header Check

4.1.4 Suspicious URL Analysis

The email contains a button labelled "Review recent activity." Examination of the HTML source showed that the button points to:

https://bit.ly/3vF9xKz

This is a shortened URL rather than a direct Microsoft website address. Shortened links can hide the final destination, making it difficult for the recipient to identify where the link actually leads. The URL was checked using VirusTotal without opening it directly. The result showed **1/91 security vendors** identifying it as phishing.

```
<p>If this was you, you can ignore this message. If this wasn't you, please secure your account immediately.</p>
<p style="text-align: center; margin: 25px 0;">
  <a href="https://bit.ly/3vF9xKz" style="background-color: #0078d4; color: white; padding: 12px 30px; text-decoration: none; border-radius: 4px; font-weight: bold;">Review recent activity</a>
</p>
```

Figure 3: Microsoft URL Check

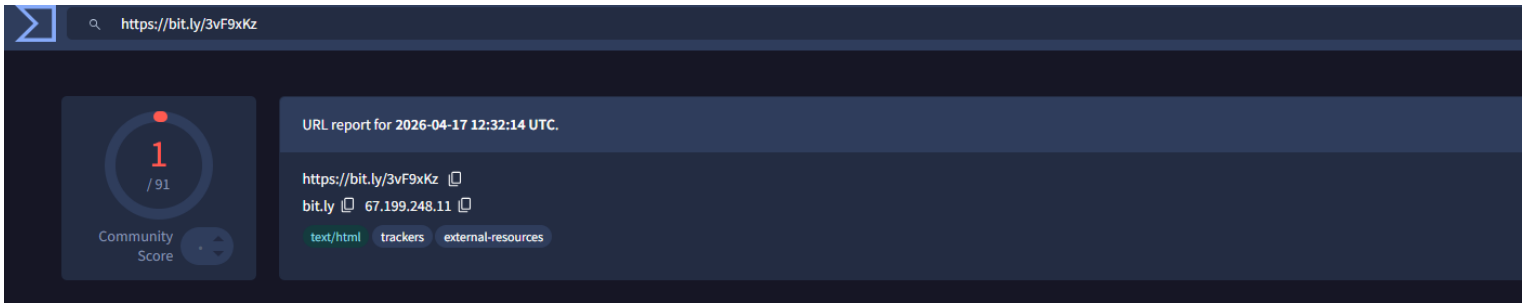


Figure 4: Microsoft Virustotal Check

4.1.5 Social Engineering Techniques

The email uses Microsoft's name to create a sense of authority. It also creates fear by suggesting that someone may have accessed the recipient's account.
The subject includes "[Action Required]" to encourage an immediate response. The attacker wants the recipient to click the link before carefully checking the sender's address or the destination.

4.1.6 Findings

This email contains several phishing warning signs, including a lookalike sender domain, failed authentication checks, a shortened URL, and an urgent account security warning.
The main lesson from this sample is that users should verify account security alerts through the company's official website or application instead of clicking links in unexpected emails.

4.2 Sample 2 – DHL Delivery Scam

4.2.1 Email Overview

The second sample is a suspicious email pretending to be from DHL. It claims that a parcel could not be delivered and asks the recipient to pay a small redelivery fee. The message warns that the parcel will be returned if no action is taken within five days.

Table 2: Sample_2 Email Detail

Field	Information
Display Name	DHL Delivery Service
Sender Address	delivery@dhl-parcel-phish-update.com
Subject	Your DHL parcel could not be delivered - action required
Date	17 October 2024

4.2.2 Sender and Domain Analysis

The sender uses the domain dhl-parcel-phish-update.com, which is not an official DHL domain. The DHL name is included to make the email appear trustworthy.
ICANN Lookup did not return a registration record for this domain, so its registration date and age could not be confirmed.

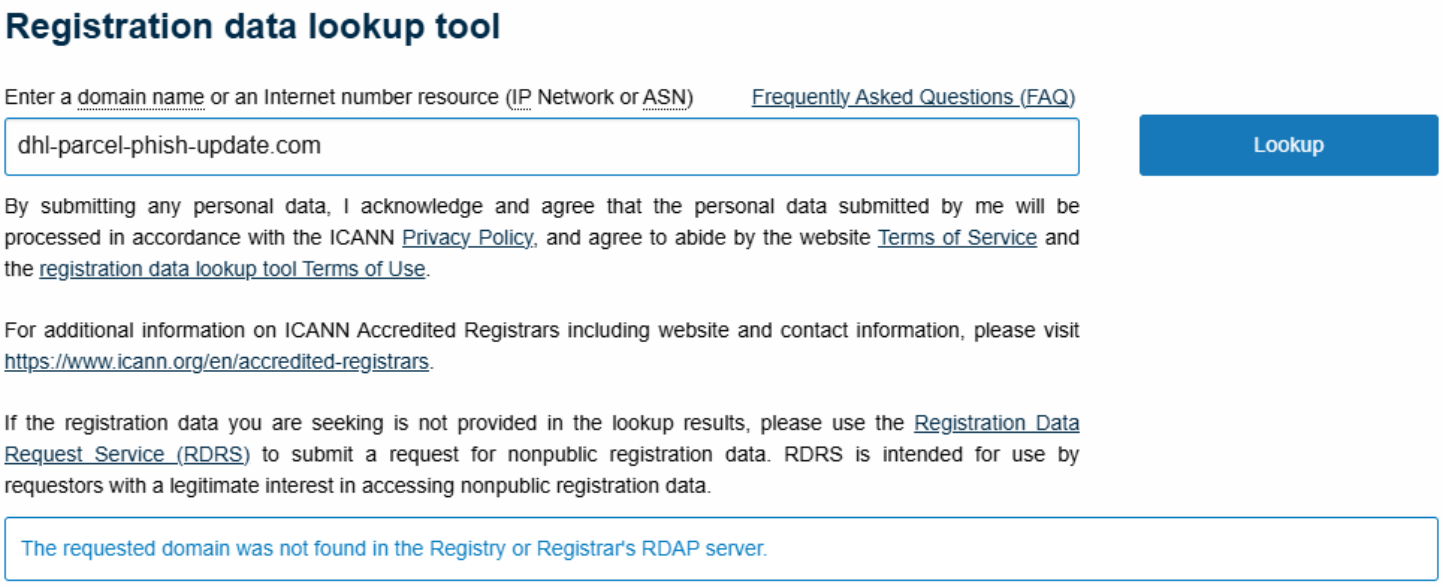


Figure 5: DHL Domain Check

4.2.3 Email Authentication Results

The email header showed the following results:

- **SPF: SOFTFAIL** – The sending server did not fully pass the SPF authorization check.
- **DKIM: NONE** – No DKIM signature was available for verification.
- **DMARC: FAIL** — The email failed the DMARC authentication check.

These results raise concerns about the authenticity of the sender.

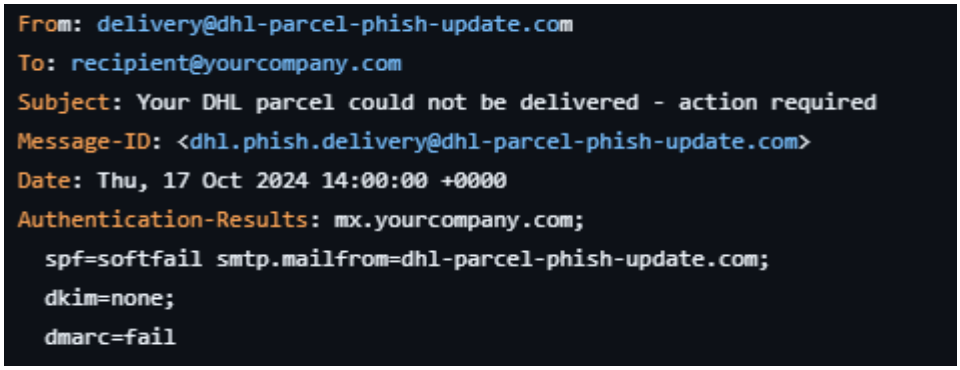


Figure 6: DHL Header Check

4.2.4 Suspicious URL Analysis

The email asks the recipient to follow a redelivery link and pay EUR 2.99 to receive the parcel.
The URL was checked using VirusTotal, which showed 0/93 security vendors flagging it. However, this does not guarantee that the link is safe.

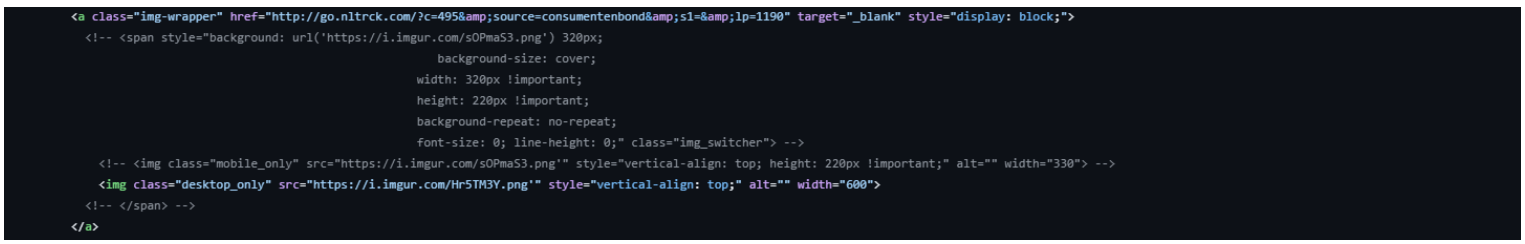


Figure 7: DHL Link Check

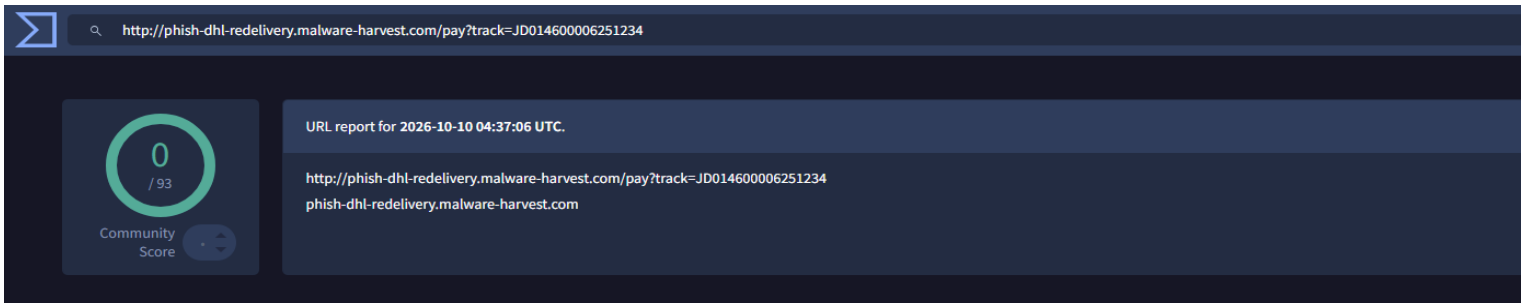


Figure 8: DHL Virustotal Check

4.2.5 Social Engineering Techniques

The email uses DHL's name to gain the recipient's trust. It creates urgency by warning that the parcel will be returned within five days.
The small payment request makes the message seem reasonable and may encourage recipients to enter their payment details without checking the website carefully.

4.2.6 Findings

The main warning signs are the suspicious sender domain, unsuccessful authentication checks, unexpected payment request, and urgency.
This sample shows how delivery-related messages can trick users into visiting suspicious payment pages. Delivery information should always be verified through the courier's official website or application.

4.3 Sample 3 – Solar Panel Promotional Email

4.3.1 Email Overview

The third sample is a Dutch-language promotional email offering solar panel quotations. It mentions rising electricity and gas prices and encourages the recipient to consider solar panels to reduce energy costs.

Table 3: Sample_3 Email Detail

Field	Information
Display Name	Zonnepanelen installateur
Sender Address	zonnepaneel@appjj.serenitepure.fr
Subject	Zonnepanelen voor een goede prijs
Date	3 November 2022
Sender IP	57.128.69.202

4.3.2 Sender and Domain Analysis

The sender uses the name "Zonnepanelen installateur" (Solar Panel Installer). However, the sender's address uses serenitepure.fr, while the Reply-To address uses aichakandisha.com and the Return-Path uses dturm.de.
These different domains make it difficult to identify the organization behind the email. The sender details also do not clearly match the advertised solar panel service.

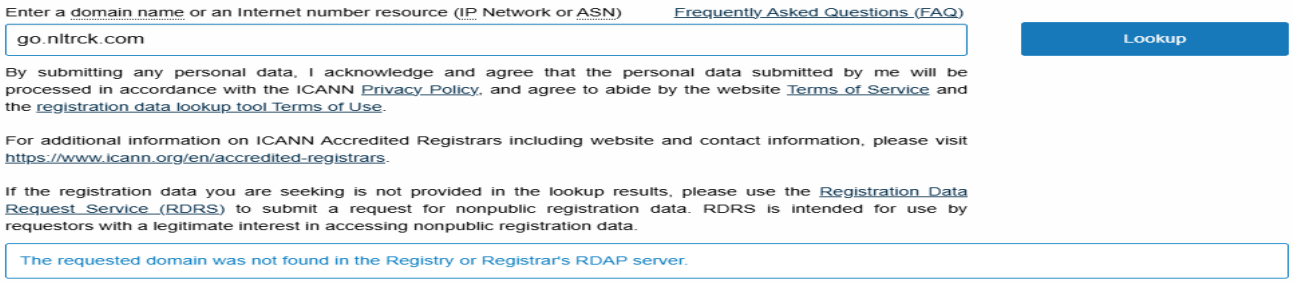


Figure 9: Solar Panel Domain Check

4.3.3 Email Authentication Results

The email header showed the following results:

- **SPF: NONE** — No SPF authentication result was established.
- **DKIM: NONE** — No DKIM signature was available for verification.
- **DMARC: NONE** — No DMARC authentication result was available.

The header also showed a composite authentication failure. These results provide limited assurance about the sender's authenticity.

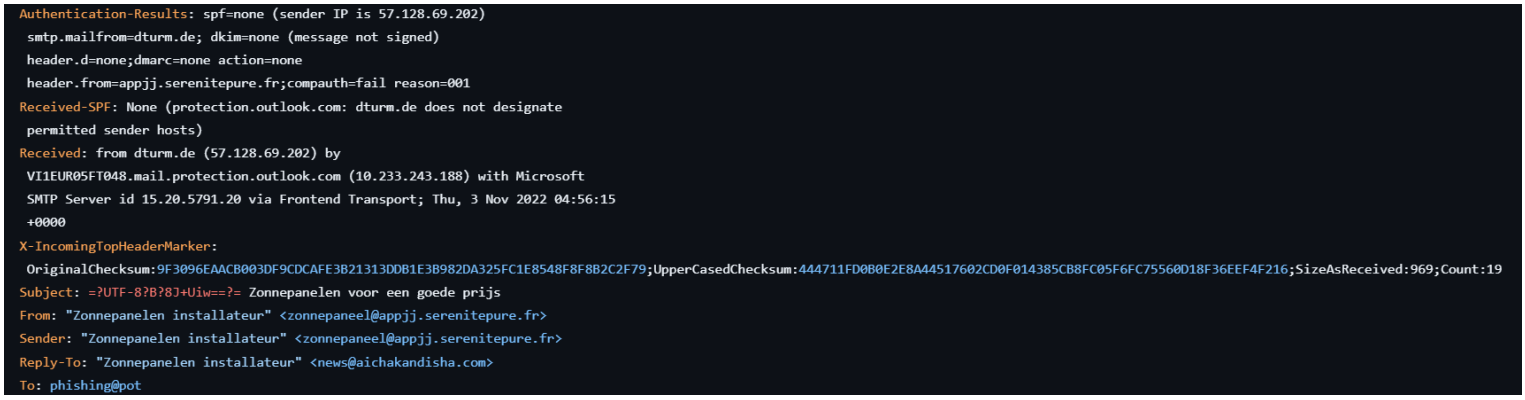


Figure 10: Solar Panel Header Check

4.3.4 Suspicious URL Analysis

The email encourages the recipient to request solar panel quotations. Examination of the email content showed the following link:

http://go.nltrck.com/?c=495&source=consumentenbond&s1=&lp=1190

The link uses nltrck.com, which differs from the sender's domain. This may indicate a separate tracking or marketing service, but the final destination cannot be confirmed from the link alone. VirusTotal showed **0/92** security vendors flagging the URL. However, this result does not guarantee that the link is safe.

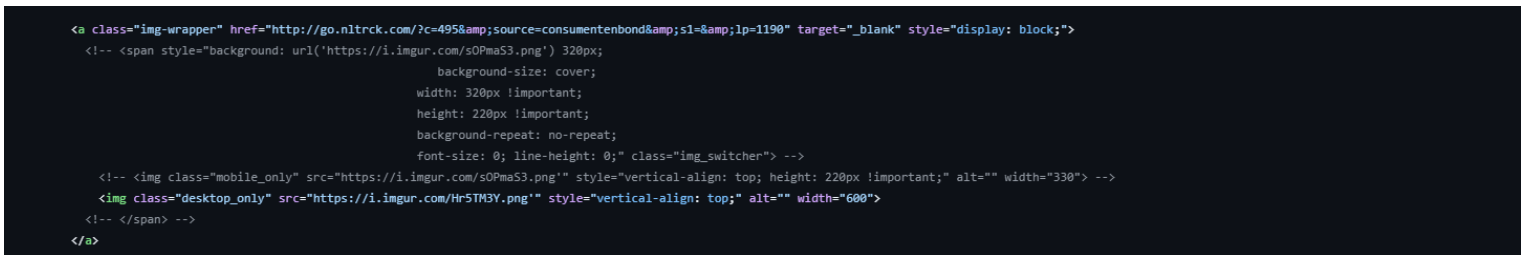


Figure 11: Solar Panel Link Check

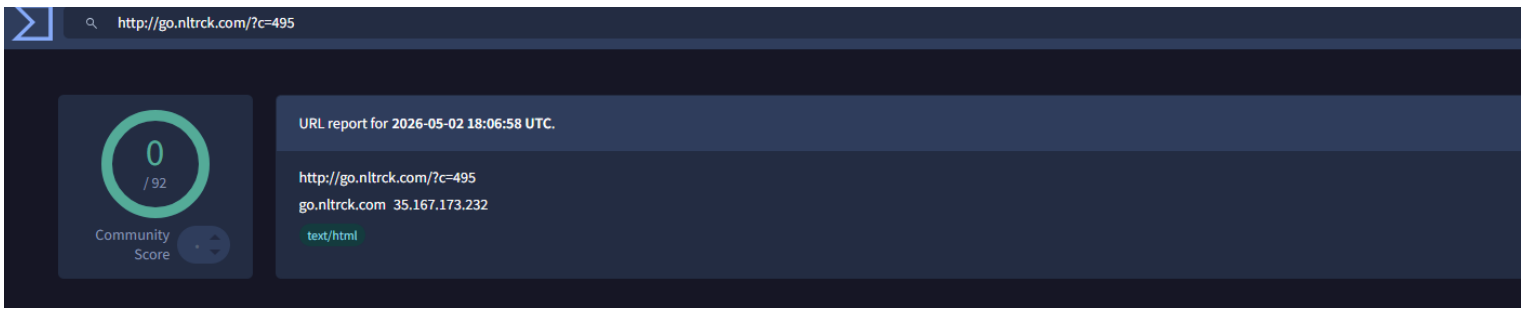


Figure 12: Solar Panel VirusTotal Check

4.3.5 Social Engineering Techniques

The email uses rising energy costs and possible financial savings to attract the recipient's attention.

It creates urgency with the message "Wacht daarom niet langer en meld u nu aan!" (Don't wait any longer and sign up now). The main techniques are financial appeal, urgency, and an attractive promotional offer.

4.3.6 Findings

The main warning signs are inconsistent sender-related domains, missing authentication results, and a link using a different domain.
This sample appears to be suspicious promotional or spam email, but the available evidence does not conclusively prove phishing. Users should verify the company and website before providing personal information.

4.4 Sample 4 – Tessa Dating Email

4.4.1 Email Overview

The fourth sample is a Dutch-language dating-related email from a sender named Tessa. The message suggests meeting in person and encourages the recipient to view an online profile. It uses a friendly and personal tone to attract attention.

Table 4: Sample_4 Email Detail

Field	Information
Display Name	Tessa
Sender Address	info@mail.befunhub.biz
Subject	zullen we afspreken?
Date	8 July 2026
Sender IP	192.99.246.208

4.4.2 Sender and Domain Analysis

The email uses the sender name "Tessa" to appear personal. The sender address is **info@mail.befunhub.biz**, while the Reply-To address is **customer@mail.befunhub.biz**. Both addresses use the same main domain.
ICANN Lookup showed that befunhub.biz was registered on **21 November 2025**. The domain was less than one year old when the email was sent. A recently registered domain is worth investigating, although its age alone does not prove malicious activity.

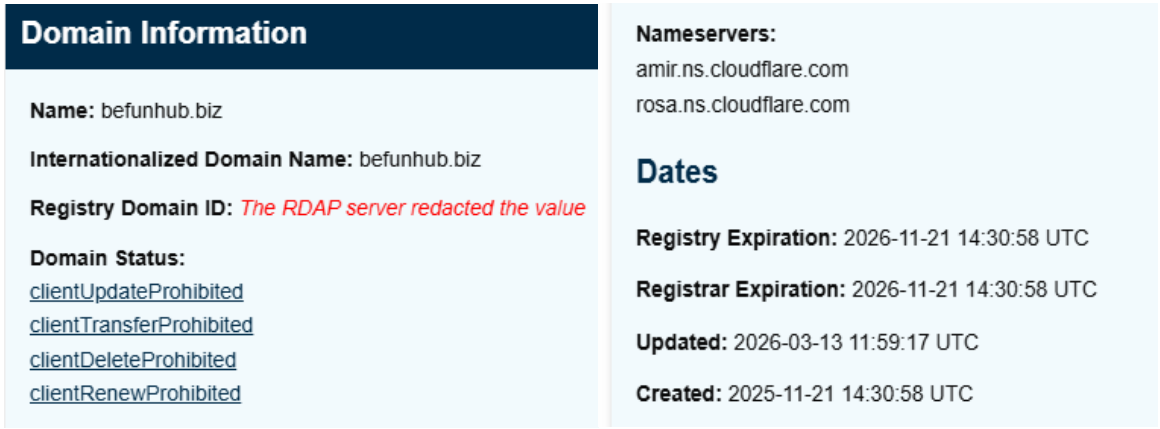


Figure 13: Tessa Domain Check

4.4.3 Email Authentication Results

The email header showed the following results:

- **SPF: PASS** — The sending server passed the SPF authorization check.
- **DKIM: PASS** — The email passed digital signature verification.
- **DMARC: PASS** — The email passed the DMARC authentication check.

All three checks passed. However, successful email authentication does not guarantee that the message itself is trustworthy.

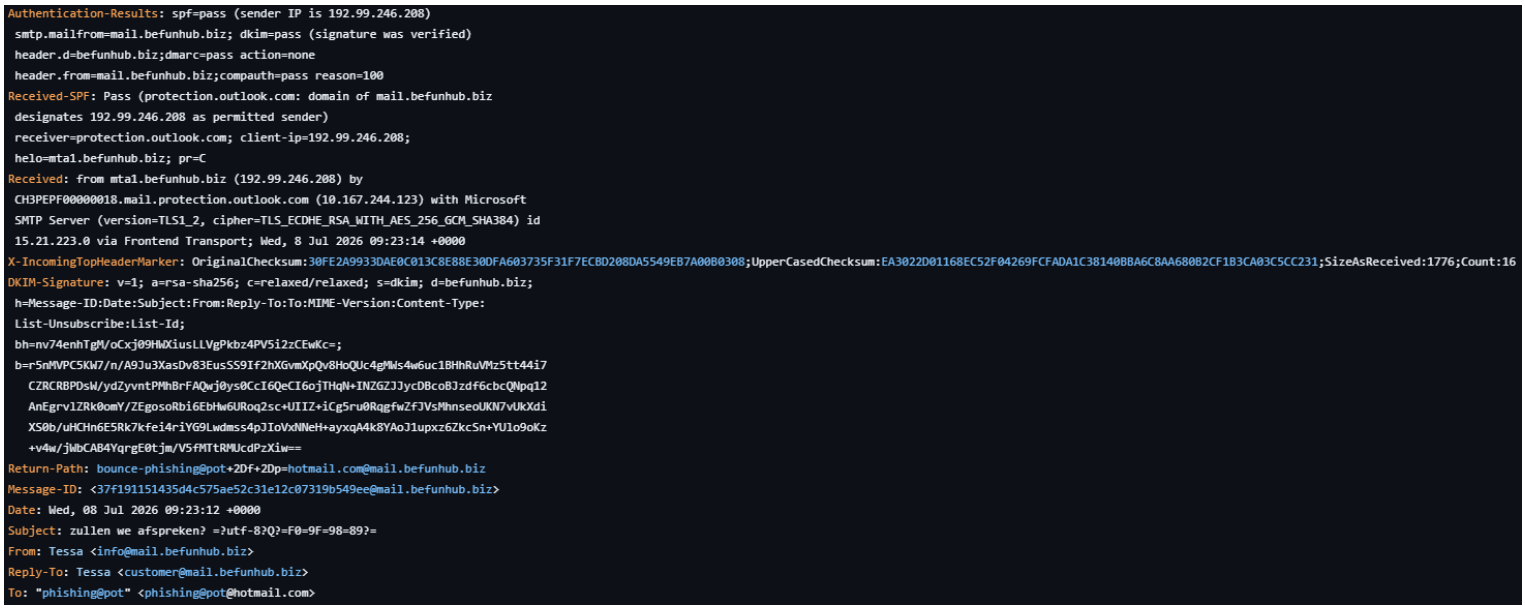


Figure 14: Tessa Header Check

4.4.4 Suspicious URL Analysis

The email contains a link encouraging the recipient to view Tessa's profile. Examination of the HTML source showed that the link uses the mail.befunhub.biz domain and contains a tracking URL. VirusTotal showed **0/93 security vendors** flagging the link. The scan also indicated that the link redirects to another website associated with an age-verification page.
This redirect is worth investigating because the destination differs from the original link. However, neither the redirect nor the zero-detection result conclusively establishes whether the website is malicious.



Figure 15: Tessa Link Check

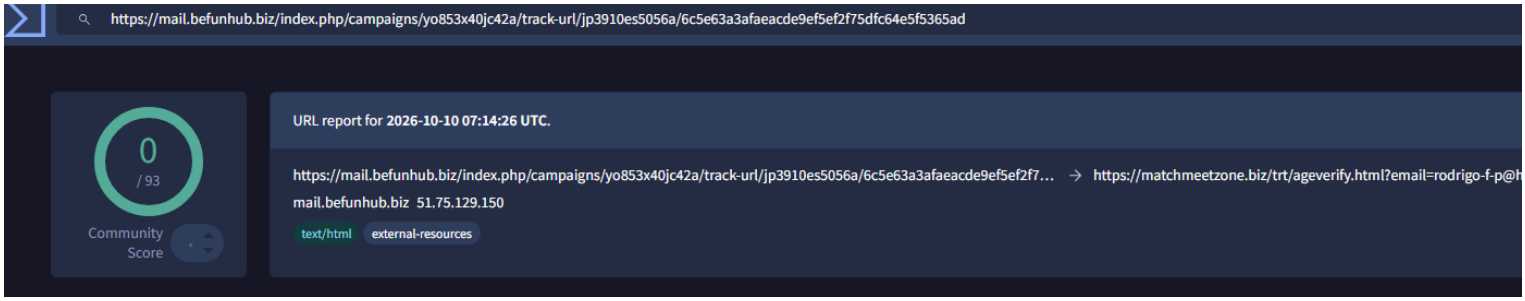


Figure 16: Tessa VirusTotal Check

4.4.5 Social Engineering Techniques

The email uses curiosity and personal interest to encourage the recipient to click the link. The suggestion of meeting someone in person makes the message feel like a private invitation. Unlike the Microsoft and DHL samples, this email does not rely mainly on fear or urgent warnings. Instead, it attempts to attract attention through a personal message.

4.4.6 Findings

The main warning signs are the unexpected dating invitation, a tracking link, and a redirect to another website.

Although SPF, DKIM, and DMARC all passed, these results only support the authenticity of the sending domain, not the safety of the email content.This sample demonstrates that suspicious emails can pass authentication checks.

4.5 Sample 5 – Amazon Account Suspension Scam

4.5.1 Email Overview

The fifth sample is a phishing email impersonating Amazon. It claims that the recipient's account has been locked because of a billing or payment information problem. The message asks the recipient to update their information immediately to avoid losing access to the account.

Table 5: Sample_5 Email Detail

Field	Information
Display Name	noreply@Quick Response
Sender Address	nuthostsrl.SaintU74045Walker@comunidadeduar.com.ar
Subject	We locked your account for security reason
Date	8 September 2023
Sender IP	40.107.215.98

4.5.2 Sender and Domain Analysis

The email claims to represent Amazon, but the actual sender address uses comunidadeduar.com.ar, which is unrelated to Amazon's official domain. The unusual sender name and long email address are additional warning signs. They do not match the identity of the company being impersonated.

An ICANN Lookup search for **comunidadeduar.com.ar** returned an error stating that the RDAP query could not be performed. Therefore, the domain's registration date and age could not be confirmed.

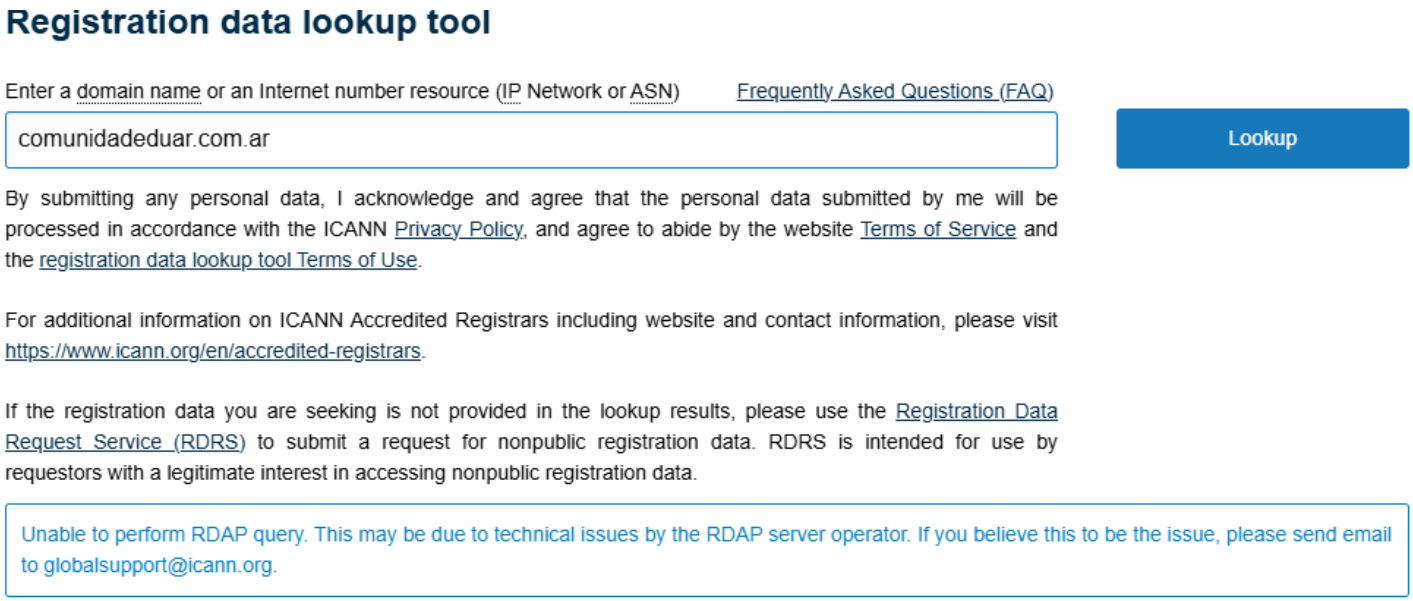


Figure 17: Amazon Domain Check

4.5.3 Email Authentication Results

The examined email headers showed:

- **SPF: PASS** — The sending server passed the SPF authorization check.
- **DKIM: NONE** — No DKIM signature was available for verification at this stage.
- **DMARC: NONE** — The original authentication results showed no DMARC result for the sender's domain.

These results show that some authentication checks were successful, but they do not establish that the email genuinely came from Amazon.



Figure 18: Amazon Header Check

4.5.4 Suspicious URL Analysis

The email contains an "Update Information" button that asks the recipient to correct their account details. Examination of the HTML source revealed the following URL:

https://script.google.com/macros/s/AKfycbwmwl6-oDDgtuO7lmFuHnvuNd mvDQlzNJaRxbTkdag0Q7lygpC3YzQqCKpTWl-aWsCqw/exec

The link uses Google Apps Script rather than an official Amazon website. Although Google provides a legitimate scripting service, this URL does not match the company the email claims to represent. VirusTotal showed **1/92 security vendors** identifying the URL as phishing. This supports the suspicion raised by the email's content and sender details.

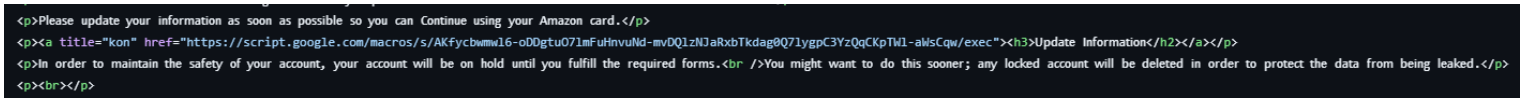


Figure 19: Amazon Link Check

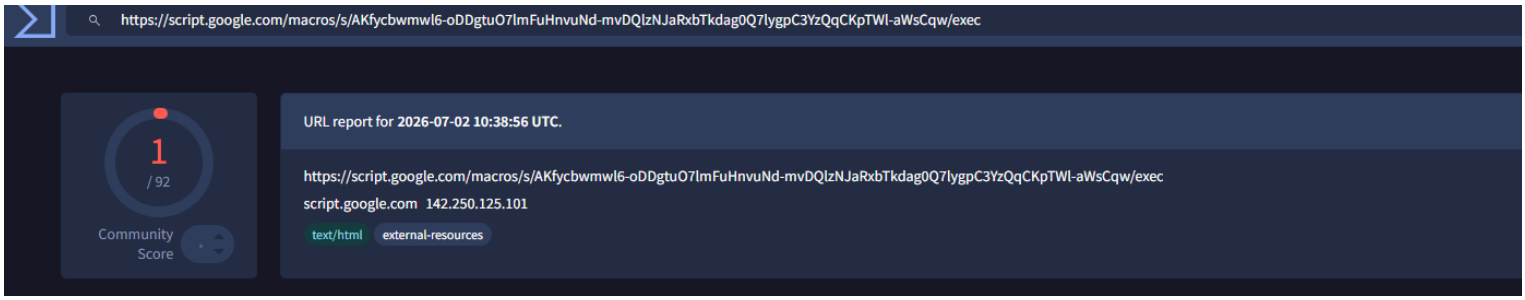


Figure 20: Amazon VirusTotal Check

4.5.5 Social Engineering Techniques

The email uses Amazon's identity to gain the recipient's trust. It creates fear by claiming that the account has been locked because of a payment information problem. It also creates urgency by warning that the account may be deleted if the recipient does not update their information. These warnings encourage the recipient to click the link without carefully checking its destination.

4.5.6 Findings

The main warning signs are the unrelated sender domain, unusual sender address, non-Amazon update link, and threats about account deletion. This sample demonstrates how attackers can combine brand impersonation, fear, and urgency

to encourage users to disclose sensitive information. Account problems should be verified through Amazon's official website or application rather than through unexpected email links.

5. Red Flag Comparison Table

The following table compares the main warning signs identified in the five email samples. It highlights differences in sender domains, authentication results, suspicious links, and social engineering techniques.

Table 6: Comparison of Phishing and Suspicious Email Samples

Indicator	Sample 1: Microsoft	Sample 2: DHL	Sample 3: Solar Panels	Sample 4: Tessa	Sample 5: Amazon
Sender domain	microsoftonline-verify.com	dhl-parcel-phish-update.com	appjj.serenitepure.fr	mail.befunhub.biz	comunidadeduar.com.ar
Domain registration	Not found	Not found	Not confirmed	21 Nov 2025	RDAP query failed
SPF	FAIL	SOFTFAIL	NONE	PASS	PASS
DKIM	FAIL	NONE	NONE	PASS	NONE
DMARC	FAIL	FAIL	NONE	PASS	NONE
Suspicious link	Shortened bit.ly URL	Redelivery payment link	nltrck.com tracking link	Tracking link with redirect	Google Apps Script link
VirusTotal	1/91	0/93	0/92	0/93	1/92
Social engineering	Fear, urgency, impersonation	Urgency, payment request, impersonation	Financial appeal, urgency	Curiosity, personal interest	Fear, urgency, impersonation
Main red flags	Lookalike domain, failed authentication	Fake delivery problem, payment request	Different sender-related domains	Unexpected invitation, redirect	Unrelated sender domain, non-Amazon link

5.1 Key Observations from the Comparison

The Microsoft, DHL, and Amazon samples use trusted company names to make their messages appear genuine. The solar panel and dating emails rely more on financial benefits and personal interest.
Authentication results also vary. Microsoft and DHL failed important authentication checks, while the Tessa email passed SPF, DKIM, and DMARC. This shows that successful authentication does not automatically mean an email is safe.
VirusTotal detection results were low for the checked links. Therefore, users should consider several warning signs together rather than relying only on a security scan result.

6. Key Findings and Recommendations

6.1 Key Findings

The analysis of five email samples showed that suspicious messages use different methods to influence recipients. The main findings are:

1. **Sender impersonation:** Some emails used familiar company names, but their actual sender domains did not match the organizations they claimed to represent.
2. **Email authentication:** SPF, DKIM, and DMARC failures were useful warning signs. However, the Tessa sample passed all three checks, showing that authentication alone cannot confirm whether an email is safe.
3. **Suspicious links:** Shortened URLs, tracking links, redirects, and links to unrelated domains made it difficult to verify the destination.
4. **Social engineering:** Fear, urgency, financial offers, and curiosity were used to encourage recipients to respond or click links.

5. **Security scan limitations:** Low VirusTotal detection counts did not necessarily mean that the links were trustworthy.

6.2 Recommendations

Based on these findings, the following precautions are recommended:

1. Check the actual sender email address instead of trusting only the displayed name.
2. Avoid clicking links or opening attachments in unexpected emails.
3. Verify account alerts, deliveries, and payment requests through official websites or applications.
4. Be cautious of messages demanding immediate action or requesting sensitive information.
5. Report suspicious emails to the organization's IT or security team.
6. Use email authentication checks and security tools as supporting evidence rather than relying on a single result.

6.3 Conclusion

This project demonstrated that phishing and suspicious emails cannot always be identified through one warning sign. A combination of sender verification, authentication checks, URL inspection, and awareness of social engineering techniques provides a more reliable way to recognize potential threats.
The five samples also showed that even emails passing authentication checks may contain suspicious content. Therefore, users should remain cautious and verify unexpected requests before taking action.

7. References

The following sources and online tools were used to collect and analyze the email samples.

7.1 Email Sample Sources

1. Jacob D. Cook – Phishing Analysis Lab (Microsoft email sample). GitHub. https://github.com/jacobdcook/Phishing-Analysis-Lab
2. Strixhack – Phishing Triage Pipeline (DHL email sample). GitHub. https://github.com/Strixhack/phishing-triage-pipeline
3. Phishing Pot – Public phishing email collection (Solar Panel and Tessa email samples). https://www.phishingpot.com/
4. Ojasy – Phishing Email Analysis (Amazon email sample). GitHub. https://github.com/ojasy/Phishing-Email-Analysis

7.2 Analysis Tools

1. ICANN Lookup – Used to examine domain registration information. https://lookup.icann.org/
2. VirusTotal – Used to check suspicious URLs against security vendor detections. https://www.virustotal.com/